

NETFLIX CONTENT VISUALIZATION

1. Introduction

Netflix is one of the world's leading streaming platforms, offering a vast collection of movies and television shows. The purpose of this project is to analyze Netflix content using data visualization techniques and gain insights into content distribution, ratings, release trends, and country-wise contributions.

2. Objectives

- To perform data cleaning and preprocessing on the Netflix dataset.
 - To analyze Netflix content using visualization techniques.
 - To identify trends in movies and TV shows.
 - To examine content ratings and country-wise distribution.
 - To create an interactive dashboard using Power BI.
-

3. Dataset Description

Dataset Name: Netflix Movies and TV Shows Dataset

Source: Kaggle

Attributes Used:

- show_id
- type
- title
- director
- cast
- country
- date_added
- release_year
- rating
- duration
- listed_in
- description

Total Records: 7,789

4. Tools and Technologies Used

- Python
 - Google Colab
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Power BI
 - GitHub
-

5. Data Cleaning

The following preprocessing steps were performed:

- Loaded the dataset into Google Colab.
- Checked for missing values.
- Removed duplicate records.
- Verified data types and column information.
- Saved the cleaned dataset for further analysis.

(Insert Screenshot: Dataset Preview)

(Insert Screenshot: Missing Values)

6. Data Visualization and Analysis

6.1 Movies vs TV Shows

A bar chart was created to compare the number of Movies and TV Shows available on Netflix.

Observation:

Movies constitute a larger portion of Netflix content compared to TV Shows.

(Insert Screenshot)

6.2 Rating Distribution

A visualization was created to analyze the distribution of content ratings.

Observation:

TV-MA and TV-14 are among the most common ratings, indicating a large amount of mature audience content.

(Insert Screenshot)

6.3 Top Countries

A country-wise analysis was performed to identify which countries contribute the most content.

Observation:

The United States contributes the highest amount of content, followed by other major content-producing countries.

(Insert Screenshot)

6.4 Content by Release Year

A trend analysis was conducted using release years.

Observation:

The number of titles increased significantly in recent years, demonstrating Netflix's growing content library.

(Insert Screenshot)

7. Power BI Dashboard

An interactive dashboard was created in Power BI containing:

- Total Content Card
- Movies Card
- TV Shows Card
- Movies vs TV Shows Chart
- Rating Distribution Chart
- Top Countries Chart
- Content by Release Year Chart
- Interactive Filters (Slicers)

(Insert Dashboard Screenshot)

8. Results and Findings

- Movies are more prevalent than TV Shows on Netflix.
- Netflix content has grown rapidly over the years.
- The United States is the leading content contributor.

- TV-MA and TV-14 ratings dominate the platform.
 - Interactive dashboards help users explore content trends efficiently.
-

9. Conclusion

This project successfully analyzed Netflix content using Python and Power BI. Various visualizations were created to understand content distribution, ratings, release trends, and country-wise contributions. The project demonstrates the practical application of data analytics and visualization techniques in extracting meaningful insights from real-world datasets.

10. References

- [Kaggle Netflix Dataset](#)
- [Python Documentation](#)
- [Power BI Documentation](#)
- [Google Colab Documentation](#)